

Safety Communication Protocols in Industrial Automation:

A Comparative Study of PROFI-safe, FSoE, CIP Safety, openSAFETY, and CC-Link IE Safety

Sumon Boiddo
Electronic Engineering
Hochschule Hamm-Lippstadt
Lippstadt, Germany
sumon.boiddo@stud.hshl.de

Nnaemeka Nnachi-Egwu
Electronic Engineering
Hochschule Hamm-Lippstadt
Lippstadt, Germany
nnaemeka.nnachi-egwu@stud.hshl.de

Oluwasholape Daniel Oyemade
Electronic Engineering
Hochschule Hamm-Lippstadt
Lippstadt, Germany
oluwasholape-daniel.oyemade@stud.hshl.de

Md Tawhidur Rahman Tuhin
Electronic Engineering
Hochschule Hamm-Lippstadt
Lippstadt, Germany
md-tawhidur-rahman.tuhin@stud.hshl.de

Abstract—Standard IT networks offer best-effort delivery with no bounded latency guarantee—a fundamental incompatibility with the demands of industrial safety systems, where a single missed or delayed packet can result in a fatal injury. This paper presents a comparative analysis of the five dominant functional-safety communication protocols deployed over Industrial Ethernet: PROFI-safe, Fail Safe over EtherCAT (FSoE), CIP Safety, openSAFETY, and CC-Link IE Safety. All five are governed by IEC 61784-3 and implement the Black Channel principle, which isolates safety logic from the transport layer by embedding independent error detection directly in each safety payload. The protocols are analysed across six dimensions: frame architecture, cyclic redundancy check scheme, passivation mechanism, real-time performance (latency, jitter, and minimum cycle time), third-party certification pathway under IEC 61508, and actual industry deployment across eight industrial sectors. Findings confirm that all five reach SIL 3 / PL e and that no single protocol is superior in every dimension. Differentiating factors include cycle-time determinism, geographic ecosystem, device catalogue breadth, and the cost and openness of the certification pathway. Protocol-selection guidance is provided for each major sector.

Index Terms—Functional safety, IEC 61508, IEC 61784-3, PROFI-safe, FSoE, CIP Safety, openSAFETY, CC-Link IE Safety, Black Channel, Industrial Ethernet, SIL

I. INTRODUCTION

Consider a straightforward question: would you use standard Wi-Fi to transmit an emergency-stop command to a 200-tonne hydraulic press? Instinct says no, and the engineering reason is precise. Standard Ethernet and Wi-Fi are designed for best-effort delivery—they maximise average throughput but make no guarantee that any individual frame will arrive within a bounded time window. For most applications this is acceptable.

For a safety circuit, a single late or lost packet can mean a robot arm strikes an operator, a press brake closes on a hand, or a conveyor continues running during maintenance.

Functional safety is defined by IEC 61508 as the part of overall safety that depends on the correct operation of electrical, electronic, and programmable systems [1]. The framework does not ask how often a hazard will occur; it asks how reliably the system responds when one does. The key metric is the Probability of dangerous Failure per Hour (PFH), and machines that guard human operators must typically meet Safety Integrity Level 3 (SIL 3), which requires $PFH < 10^{-7}/h$. IEC 62061 [2] and ISO 13849 [3] translate these requirements into concrete design obligations—certified communication links are mandatory for Category 3 / PL d and above machine designs.

When legacy fieldbus systems gave way to Industrial Ethernet in the early 2000s, the industry gained bandwidth, flexibility, and IT compatibility, but lost the deterministic guarantees of dedicated wiring [4], [5]. Each major Industrial Ethernet ecosystem—PROFINET, EtherCAT, EtherNet/IP, POWERLINK, and CC-Link IE—subsequently developed its own safety sublayer. All of them independently arrived at the same architectural solution: treat the transport as an untrusted channel and embed all necessary error protection directly in the safety payload. IEC 61784-3 [6], first published in 2010 and significantly revised in 2021, formalised this into the *Black Channel* principle and established a unified certification framework for all safety protocol families.

The resulting market is fragmented along regional and ecosystem lines in a way that creates a practical challenge for

industry. A Siemens SIMATIC S7-1500F safety programmable logic controller ships on identical hardware worldwide, yet runs PROFIsafe on European production lines and CIP Safety in North American facilities. The protocol is not determined by the hardware manufacturer but chosen by the system integrator based on the existing network infrastructure and regional supply chains. A safety engineer who works across multiple sites must therefore be competent in several protocols simultaneously.

This paper provides a structured comparative analysis of the five protocols that together cover more than 95% of certified safe nodes currently deployed on Industrial Ethernet [6], [7]. Each is examined on its frame architecture, error-detection scheme, passivation behaviour, performance characteristics, certification pathway, and real-world deployment across major industrial sectors. The paper is organised as follows: Section II reviews the evolution of industrial safety communication and the three channel architectures defined by IEC 61784-3. Section III describes and compares the architecture of each protocol. Section IV evaluates performance, certification, and industry deployment. Section V presents conclusions and protocol-selection guidance.

II. BACKGROUND

A. Evolution of Industrial Safety Communication

The history of industrial safety communication begins in the 1980s with dedicated fieldbus systems. PROFIBUS DP, DeviceNet, and CC-Link provided deterministic, low-latency communication over proprietary physical layers; safety was implemented through dedicated wiring, relay circuits, or vendor-specific sublayers [8]. During this period, CANopen Safety (CiA 304) [9] introduced the concept of embedding safety data within a standard frame on CAN-bus networks—a technique later generalised into the Black Channel model. CANopen Safety remains relevant today in medical devices and building automation, but its reliance on CAN-bus speeds and topologies places it outside the scope of this Industrial Ethernet comparison.

The shift to Industrial Ethernet in the early 2000s brought dramatic performance improvements: gigabit speeds, standard RJ-45 cabling, and compatibility with enterprise IT infrastructure [5]. However, standard Ethernet provides no timing guarantees, and each Industrial Ethernet variant solved real-time requirements differently. EtherCAT processes frames on the fly as they pass through each node; PROFINET uses scheduled isochronous real-time (IRT) slots; EtherNet/IP relies on UDP-based cyclic messaging. Safety extensions followed between 2005 and 2012, one per ecosystem, all independently converging on the same approach. IEC 61784-3 unified their certification requirements under the Communication Profile Family (CPF) model, assigning each protocol a unique identifier [6].

B. Black, White, and Grey Channel Architectures

IEC 61784-3 identifies three architectures for safety communication, distinguished by the level of trust placed in the

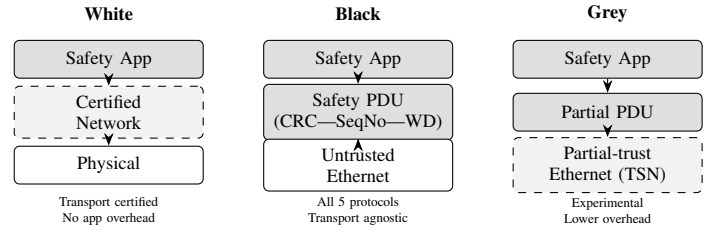


Fig. 1. The three IEC 61784-3 channel architectures. White Channel certifies the transport itself; Black Channel treats the transport as fully untrusted; Grey Channel partially relies on transport-level safety functions. All five protocols in this paper use Black Channel.

transport layer [6], as illustrated in Fig. 1.

In a *White Channel*, the transport is fully certified as safe. The application layer requires no additional protective measures because the network guarantees error-free, bounded delivery by design. Historical examples include SafetyBUS p and fieldbus-level AS-i Safety. While conceptually simple, the White Channel demands certification of the entire network infrastructure, making new deployments expensive, inflexible, and largely obsolete.

In a *Black Channel*, the transport is treated as completely untrusted—equivalent to an adversarial or unreliable channel. The safety protocol adds its own cyclic redundancy check (CRC), consecutive sequence number, connection identifier, and watchdog timer directly to each payload frame. Together these measures detect all seven transmission error classes defined by IEC 61784-3: corruption, unintended repetition, incorrect sequence, loss, unacceptable delay, insertion, and masquerading. No assumptions are made about the underlying network, meaning the same safety sublayer can run unchanged over PROFINET, Wi-Fi, or Modbus TCP—a significant advantage for system integrators.

A *Grey Channel* occupies intermediate ground: the transport provides some safety functions, such as CRC or sequence numbering built into the network protocol itself, but is not fully certified. The safety application builds only the remaining required measures on top, reducing overhead compared to the Black Channel. This model is currently being explored in Time-Sensitive Networking (TSN) profiles under IEC/IEEE 60802 but has not yet reached the standardisation maturity of the Black Channel approach.

C. Governing Standards

The regulatory framework governing functional-safety communication operates at three levels. IEC 61508 [1] is the foundation, defining SIL 1–4, PFH targets, and both hardware and software development lifecycle requirements applicable to all safety-related systems. For machine safety specifically, IEC 62061 [2] defines SIL requirements for Safety-Related Control Functions (SRCF) and ISO 13849-1 [3] provides the parallel Performance Level (PL a–e) framework widely used by European machine builders. IEC 61784-3 [6] governs communication directly, assigning each protocol a Communication Profile Family identifier: CP 3/1 for PROFIsafe, CP 12/1 for

FSoE, CP 2/1 for CIP Safety, CP 13/1 for openSAFETY, and CP 8/1 for CC-Link IE Safety. Installations in the process industry also reference IEC 61511, which extends IEC 61508 to safety instrumented systems in oil, gas, and chemical plant environments.

III. PROTOCOL ARCHITECTURES

A. PROFIsafe (CP 3/1)

PROFIsafe [6], [7] is built on an F-Host/F-Device model in which a safety PDU (F-PDU) is inserted transparently into standard PROFINET IO data slots, requiring no modification to the underlying PROFINET infrastructure. Each F-PDU carries a CRC-24 checksum and an 8-bit F-Address combined with a monotonically increasing consecutive number, which together provide sequence integrity and node identification. The watchdog timeout is configured per-device through the `F_WD_Time` parameter; expiry causes the F-Supervisor within the F-Host to passivate all outputs of the affected F-Device. The engineering environment is TIA Portal Safety, and the certified device catalogue exceeds 2000 products.

The F-PDU supports payloads of up to 123 bytes, accommodating complex safety functions such as safe-torque-off (STO) commands for multiple axes in a single frame. PROFIsafe operates over both PROFINET RT (software real-time) and PROFINET IRT (isochronous real-time); the latter provides sub-microsecond jitter through hardware-scheduled slots.

With more than 35 million certified safe nodes globally, PROFIsafe holds the largest installed base of any functional-safety protocol. It is the dominant standard in European automotive assembly (Volkswagen Group, BMW Group, Mercedes-Benz), heavy industry (ThyssenKrupp steel mills), and process industries (BASF chemical plants, Shell refineries), where Siemens SIMATIC infrastructure is widespread.

B. Fail Safe over EtherCAT – FSoE (CP 12/1)

FSoE [6], [10] uses a Master/Slave model. Each safety frame consists of a Command Byte, User Data, Connection ID, and CRC—CRC-8 for payloads of up to two bytes, and CRC-16 for larger ones. A 2-byte Safety Header Block (SHB) carries the sequence number. A critical design feature distinguishes FSoE from the other four protocols: slave devices passivate autonomously when the watchdog expires, without waiting for a reset command from the master. This eliminates one round-trip from the reaction path and makes FSoE behaviour well-defined even in the event of master failure. Combined with EtherCAT's distributed clock synchronisation—which aligns all nodes to within 1 ns—FSoE achieves minimum cycle times of approximately 125 μ s and jitter below 1 μ s, the best figures in this comparison. The safety overhead is the most compact at approximately 6 bytes, leaving the bulk of the EtherCAT frame available for process data.

These characteristics make FSoE the protocol of choice for high-axis-count motion control: FANUC robot cells in automotive tier-1 suppliers, TRUMPF laser-cutting machines,

and Beckhoff-equipped high-speed packaging lines (Tetra Pak, Bosch Packaging Technology) all run FSoE.

C. CIP Safety (CP 2/1)

CIP Safety [6], [11] maps safety connections over the standard EtherNet/IP object model using Originator and Target roles. Its most distinctive mechanism is a dual-packet format: the actual process data and its bitwise complement are transmitted in the same frame, providing a hardware-independent cross-check that is particularly robust against systematic failures in the application layer. A Time Coordination Object compensates for variable network propagation delays, an important provision given that EtherNet/IP runs over unscheduled UDP. The Safety Supervisor Object (class 0x39) manages the connection state machine and passivation sequence. The Target enters the safe state autonomously; no command from the Originator is required.

The dual-packet design results in the highest protocol overhead among the five at approximately 12 bytes, but it also enables the deepest integration with existing EtherNet/IP infrastructure—safety and standard devices share the same network without gateway hardware. CIP Safety is the de-facto standard in North American manufacturing: General Motors stamping and body plants, Procter & Gamble consumer goods lines, and Boeing aerospace assembly all rely on it.

D. openSAFETY (CP 13/1)

openSAFETY [6], [12] employs a frame-in-frame design: the safety frame is embedded within the payload of a standard POWERLINK or Modbus TCP frame, and the transport layer processes it as ordinary data. Each Safety Process Data Object (SPDO) comprises two independent sub-frames, each protected by its own CRC (CRC-8 and CRC-16 respectively). Passivation is triggered only when both sub-frame checks fail simultaneously, which significantly reduces nuisance trips caused by transient single-bit errors—a practical advantage in electrically noisy factory environments.

openSAFETY is the only royalty-free, fully open specification among the five. Any vendor may implement the stack and submit it for independent third-party certification without licensing from a standards consortium, lowering the market-entry barrier for new device manufacturers. This openness has driven adoption among European OEM machine builders: Engel injection-moulding machines, Krones beverage bottling and labelling lines, and B&R-equipped production cells across the European packaging and printing industries all use openSAFETY.

E. CC-Link IE Safety (CP 8/1)

CC-Link IE Safety [6], [13] maps a Safety Communication Layer (SCL) into the cyclic frame structure of CC-Link IE Field. Each safety frame carries a CCITT CRC-16 checksum and combines a node address with a frame counter for sequencing and anti-replay protection. Unlike the other four protocols, which support arbitrary network topologies, CC-Link IE Safety is constrained to ring and line configurations.

Passivation is managed centrally by a Mitsubishi MELSEC iQ-R safety PLC, which monitors all nodes and coordinates the safe-state transition. The engineering environment is GX Works3.

Certified under JIS B 9961 (internationally recognised), CC-Link IE Safety is effectively mandatory for Tier 1 and Tier 2 suppliers to Japanese automotive OEMs. Toyota Motor Corporation assembly plants, Denso automotive component manufacturing, and Panasonic electronics production are representative deployments.

F. Comparative Architecture

Table I summarises the key parameters across all five protocols. Note that all five use the Black Channel model; the protocols differ in CRC polynomial, sequencing mechanism, maximum payload, and overhead, but not in their fundamental approach to safety independence from the transport layer.

TABLE I
PROTOCOL ARCHITECTURE COMPARISON [6]

Feature	PROFIsafe	FSoE	CIP	openSFTY	CC-Lnk
CPF	3/1	12/1	2/1	13/1	8/1
Channel	Black	Black	Black	Black	Black
Max payload (B)	123	126	Variable	8 (SPDO)	8
CRC	24-bit	8/16-bit	S1 (16-b)	8+16-bit	16-bit
Safety OH (B)	~4	~6	~12	~11	~6
Topology	Any	Any	Any	Any	Ring
Open spec.	No	No	No	Yes	No

IV. EVALUATION

A. Performance Metrics

Four metrics characterise real-time safety communication performance. *Minimum cycle time* is the fastest interval at which safety data can be refreshed end-to-end. *Typical jitter* measures the cycle-to-cycle variance in delivery time, which is decisive for motion-control applications where positional errors accumulate with each missed synchronisation window. *Safety reaction time* is the worst-case elapsed duration from hazard detection to safe-state actuation; it is typically two to three times the cycle time because the watchdog must expire over multiple missed cycles before triggering passivation. *PFH* is the Probability of dangerous Failure per Hour [1], the quantitative SIL metric.

Throughput is deliberately excluded from this comparison. For safety protocols, effective data rate is the product of payload size and cycle frequency; since both quantities are already captured by the cycle time and overhead rows, raw throughput adds no discriminating information. Furthermore, all five protocols operate over 100 Mbps or 1 Gbps Industrial Ethernet infrastructure, meaning the channel itself is never the bottleneck. The binding constraint in every safety application is timing, not bandwidth.

TABLE II
PERFORMANCE COMPARISON

Metric	PROFIsafe	FSoE	CIP	openSFTY	CC-Lnk
Min. cycle	~1 ms	~125 μ s	~1–10 ms	~400 μ s	~500 μ s
Jitter	<1 μ s	<1 μ s	~100 μ s	<1 μ s	~1 μ s
Reaction	~4–8 ms	~1–2 ms	~8–20 ms	~2–4 ms	~4–6 ms
Max SIL	SIL 3	SIL 3	SIL 3	SIL 3	SIL 3
PFH ceil.	< 10 ⁻⁸ /h	< 10 ⁻⁸ /h	< 10 ⁻⁸ /h	< 10 ⁻⁸ /h	< 10 ⁻⁸ /h

Sources: [7], [10], [13]–[15]

B. Performance Results

Table II presents performance data drawn from protocol specifications and vendor documentation.

FSoE achieves the lowest cycle time and jitter by a significant margin, a direct consequence of EtherCAT’s hardware-level distributed-clock synchronisation and FSoE’s autonomous slave passivation. openSAFETY follows closely, benefiting from POWERLINK’s isochronous scheduling. PROFIsafe with PROFINET IRT also achieves sub-microsecond jitter but requires a managed switch infrastructure and IEEE 1588 time synchronisation to do so. CIP Safety over EtherNet/IP operates over unscheduled UDP, which produces higher and more variable jitter; this is the trade-off for its superior interoperability with standard network equipment and its ability to share the same IP network with non-safety devices. CC-Link IE Safety occupies the middle ground, with determinism comparable to PROFIsafe IRT but constrained to ring topologies. For applications where process safety times exceed 10 ms—the majority of light-curtain and E-stop applications—all five protocols provide adequate headroom; for synchronised servo control requiring reaction times below 2 ms, FSoE is the only viable choice from this group.

C. Certification and Reliability

All five protocols require assessment by an accredited third-party certification body—TÜV SÜD, TÜV Rheinland, SGS-TÜV, or equivalent—under IEC 61508 Part 3 [1]. Hardware certification encompasses failure mode and effects analysis (FMEA), diagnostic coverage (DC) analysis, common-cause failure (CCF) evaluation, and verification that hardware fault tolerance (HFT) matches the target SIL. Software must comply with the IEC 61508 Part 3 development lifecycle; MISRA C is the prevalent coding standard for embedded safety stacks. Ongoing reliability obligations include achieving the rated mean time between failures (MTBF) and conducting periodic proof tests; SIL 3 installations typically specify proof-test intervals of one to twenty years depending on the PFH of the specific device.

openSAFETY carries a practical advantage in this dimension: as the only royalty-free open specification [12], any manufacturer may implement and independently certify the stack without paying licensing fees to a standards consortium, reducing the market-entry cost for new device vendors. The other four protocols require a licensing relationship with their respective owning organisations. CC-Link IE Safety presents

a different consideration: its certificates are issued under JIS B 9961 [13], which is internationally recognised but requires an additional administrative layer for non-Japanese customers comparing it against IEC 61508 documentation.

D. Industry Deployment

Table III maps the five protocols across eight major industrial sectors, categorising adoption as Primary (P), Common (C), or negligible (-), with representative real-world users.

TABLE III
INDUSTRY DEPLOYMENT BY SECTOR

Sector	PROFIsafe	FSoE	CIP	openSFTY	CC-Lnk
Auto. EU	P (VW, BMW)	C	C	–	–
Auto. N.Am.	C	–	P (GM,Ford)	–	–
Auto. APAC	C	C	–	–	P (Toyota)
Process ind.	P (BASF)	–	C	–	–
Machine bldg.	P	C	–	C (Engel)	–
Food & Bev.	C	C (Tetra Pak)	C	C	–
Packaging	C	P (Bosch)	C	P (Krones)	–
Electronics	C	P	C	–	C (Panasonic)

P=Primary; **C**=Common; **–**=negligible

The deployment data reinforce two conclusions. First, no protocol dominates across all sectors: PROFIsafe leads in European automotive and process industries, CIP Safety leads in North America, and CC-Link IE Safety is paramount in the APAC automotive supply chain. Second, FSoE and openSAFETY are less regionally constrained and more application-constrained: FSoE is the natural choice wherever EtherCAT motion control is already installed, while openSAFETY is favoured by machine builders seeking vendor independence.

V. CONCLUSIONS

This paper has compared PROFIsafe, FSoE, CIP Safety, openSAFETY, and CC-Link IE Safety across architecture, performance, certification, and industrial deployment. The principal findings are as follows.

All five protocols achieve SIL 3 / PL e under IEC 61508 and IEC 62061; the safety ceiling itself is not a basis for selection. The protocols differ meaningfully in cycle-time performance: FSoE and openSAFETY achieve sub-millisecond cycle times and sub-microsecond jitter, making them the preferred choice for motion-control applications; CIP Safety accepts higher jitter in exchange for seamless integration with existing EtherNet/IP infrastructure. PROFIsafe commands the largest installed base and the broadest certified device ecosystem, particularly in Europe and the process industries. openSAFETY is the only royalty-free open specification, offering the lowest certification barrier for new vendors. CC-Link IE Safety is effectively mandatory in the Japanese and broader APAC automotive supply chain.

The Black Channel architecture underpins all five protocols and represents the mature engineering consensus for functional safety over standard Industrial Ethernet. The White Channel is obsolete for new designs. The Grey Channel—currently being developed within TSN profiles under IEC/IEEE 60802—may

reduce protocol overhead while maintaining SIL 3 assurance, but has not yet reached production standardisation.

For practitioners, three factors should drive protocol selection: the transport already deployed at the target site, the regional device ecosystem and supply-chain availability, and the required safety reaction time. Engineers working globally must expect to encounter all five protocols across different sites, even on identical hardware platforms. Future work should investigate the impact of IEEE 802.1 TSN on worst-case latency guarantees for Black Channel safety sublayers, and the potential of Grey Channel architectures to reduce overhead without sacrificing SIL 3 assurance.

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